The reaction field, not the dielectric constant: solvent effects on the Menshutkin reaction $H_3N + CH_3Cl$

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November 2026

1 Introduction

In a Menshutkin reaction an amine attacks an alkyl halide to give a quaternary ammonium salt [1]. Unlike a conventional S_N2, it starts from two neutral molecules and ends in two ions, so charge is not displaced but *created* along the reaction coordinate. The Hughes–Ingold rules [2] therefore predict a large acceleration by polar solvents, and experiment confirms it: the rate spans more than four orders of magnitude from hydrocarbons to dipolar aprotic solvents [3].

The exercise reported here [4] asks for the activation barrier, the transition state (TS) geometry and the charge on the leaving chloride to be correlated with the dielectric constant ε of the medium. However, ε is a macroscopic property of the bulk liquid and does not appear on its own in any expression for the energy of the solute. In a continuum model the solvent is a structureless dielectric in which the solute carves a cavity; the solute polarises the medium, and the polarised medium acts back on it through a reaction field [5]. For the analytically solvable case of a point dipole in a spherical cavity, Onsager [6] obtained $\Delta G_{\mathrm{solv}} = -(\mu^2/a^3)\, f(\varepsilon)$ with $f(\varepsilon) = (\varepsilon - 1)/(2\varepsilon + 1)$, so that ε enters only inside a *bounded* function, $0 \le f < 1/2$; the Born and Kirkwood expressions share this feature [7]. The mapping $\varepsilon \to f(\varepsilon)$ therefore compresses the polar end of the scale severely: acetonitrile and water differ by 43 units in ε but by 0.011 in $f(\varepsilon)$.

The hypothesis of this work is that the apparent saturation of solvent effects with ε is an artefact of the chosen variable: plotted against $f(\varepsilon)$, the same quantities should be linear, and the resulting straight line should have predictive value for any solvent of known ε. The Onsager expression is used here to identify the variable, not to compute anything.

2 Methods

All calculations used Gaussian 09 [8] at the HF/6-31+G* level, the diffuse functions being required by the chloride anion and by the charge separation in the transition state (TS). Solvation was described with the polarizable continuum model [5], in which the solute polarises a structureless dielectric whose induced surface charges act back on it. SMD [9] was deliberately avoided: it adds terms parametrised from solvent descriptors other than ε, which would confound a test of whether a single electrostatic variable governs the solvent dependence. Heptane, benzene, chloroform, acetonitrile and water were used, with the gas phase as the $\varepsilon = 1$ limit fixing the origin of the $f(\varepsilon)$ axis; the dielectric constants are those applied internally by the program, not the nominal values of the handout. The reactant complex, the TS, the reactants NH_3 and CH_3Cl and the products $CH_3NH_3^+$ and Cl^- were optimized in each medium and characterised by harmonic frequencies, chained through the six media from the converged geometry of the previous one; every minimum gave real frequencies only and every TS exactly one imaginary frequency. Barriers are referred to the separated reactants, the only reference available in all six media and the one matching the second-order rate constants measured experimentally.

Atomic charges are not observables, and any partition of a continuous density among atoms is a convention. Mulliken populations [10] assign each basis function to the atom on which it is centred, which with diffuse functions no longer reflects where the density is; charges were therefore obtained with ChelpG [11], a fit to the molecular electrostatic potential, the same quantity that generates the reaction field. Relative rates follow from $k_{\mathrm{rel}} = \exp(-\Delta\Delta G^{\ddagger}/RT)$ at 298.15 K; data were extracted from the output files with an `awk` script.

3 Results and discussion

The effect does not saturate; the axis does. Correlated against ε, as the exercise requests, none of the six quantities in Table 1 is linear: R^2 lies between 0.45 and 0.58, and the barrier shows the familiar saturation above $\varepsilon \approx 30$ (Figure 1a). Against $f(\varepsilon)$, every one of them becomes a straight line, $0.977 \le R^2 \le 0.9998$. Same points, same calculations: only the abscissa changes.

A predictive model. The five solvents define $\Delta G^{\ddagger} = -32.2\, f(\varepsilon) + 43.6$ kcal/mol ($R^2 = 0.996$), which gives the barrier of any solvent from its dielectric constant alone to within 1.5 kcal/mol in a leave-one-out test. The reactant complex is not an alternative reference: the dipole–dipole interaction holding it is screened by the medium, and beyond benzene the surface is flat; where it does exist it lies *above* the separated reactants in free energy.

An earlier, less polarised transition state. Both q(Cl) and μ *decrease* with solvent polarity: the TS is less polarised in water than in vacuum, the opposite of what direct polarisation would give. The cause is

*Declaration of AI use: An AI assistant (Claude Opus 5.5, Anthropic) was consulted for editorial assistance to refine English phrasing and grammar, as well as to assist in writing a bash script for log file parsing and generating figure/table plotting code. All calculations, data processing, and intellectual content are the author's sole responsibility.

geometric. The reaction yields two separated ions and is endergonic by 114.09 kcal mol^{-1} in the gas phase, but solvation of those ions makes it exergonic by acetonitrile (Figure 1d), a swing of 129 kcal mol^{-1}. Hammond's postulate then demands an earlier TS, and d(N–C) lengthens from 1.899 to 2.209 Å while d(C–Cl) shortens from 2.474 to 2.289 Å (Figure 1c); an earlier TS has transferred less charge. The geometric displacement overrides the direct polarisation. The Leffler coefficient $\alpha = \partial\Delta G^{\ddagger}/\partial\Delta G_{\mathrm{r}} = 0.15$ confirms a reactant-like TS, and the reaction turns exergonic above $\varepsilon \approx 8$, consistent with these alkylations being run in acetonitrile or DMSO rather than in hydrocarbons.

One variable, five correlations. These correlations are not independent. The only solvent property entering a continuum calculation is ε, and only through $f(\varepsilon)$; if every property of the solute is linear in that single variable, each is linear in the others. The Hammond ($R^2 = 0.996$) and Leffler ($R^2 = 0.9995$) relations follow from that common dependence rather than testing it, and only their slopes carry independent information. The conclusion does not depend on the charge partition scheme: Mulliken charges differ in absolute value, -0.700 against -0.790 in the gas phase, but show the same trend and the same contrast, $R^2 = 0.62$ against ε and 0.99 against $f(\varepsilon)$.

Table 1: TS geometry, ChelpG charge on chlorine, dipole moment, barrier (from the separated reactants) and reaction free energy. The last two rows give R^2 of a linear fit of each column against ε and against $f(\varepsilon)$.

Medium	ε	$f(\varepsilon)$	d(N–C) Å	d(C–Cl) Å	q(Cl) e	μ D	$\Delta G^{\ddagger}$ kcal mol^{-1}	ΔG_{r} kcal mol^{-1}
Gas	1.0000	0.000	1.899	2.474	−0.790	13.51	47.67	114.09
Heptane	1.9113	0.189	2.035	2.396	−0.774	12.89	37.78	51.43
Benzene	2.2706	0.229	2.061	2.381	−0.771	12.77	36.11	40.59
Chloroform	4.7113	0.356	2.136	2.335	−0.761	12.37	31.68	10.85
Acetonitrile	35.6880	0.479	2.203	2.292	−0.749	12.00	28.24	−12.97
Water	78.3553	0.490	2.209	2.289	−0.748	11.96	27.97	−14.93
R^2 vs. ε	–	–	0.510	0.527	0.576	0.549	0.447	0.464
R^2 vs. $f(\varepsilon)$	–	–	0.995	0.998	0.999	1.000	0.977	0.983

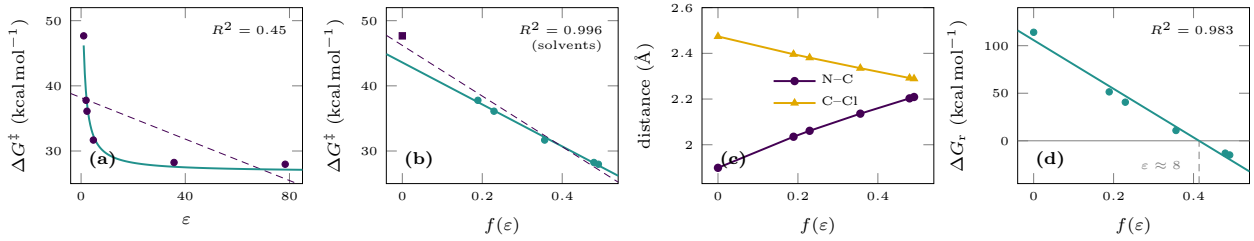


Figure 1: (a) Barrier against ε: a linear fit fails ($R^2 = 0.45$); the solid line is the model of (b) mapped onto this axis. (b) The same barriers against $f(\varepsilon)$; dashed, the fit including the gas phase. (c) TS bond distances and (d) reaction free energy against $f(\varepsilon)$.

4 Conclusion

Correlating against ε, as requested, gives no linear relationship for any of the six quantities studied ($0.45 \leq R^2 \leq 0.58$); against $f(\varepsilon)$ all of them are linear ($R^2 \geq 0.977$). The apparent saturation of solvent effects is a property of the axis, not of the chemistry, and the resulting line predicts the barrier of any solvent from ε alone to within 1.5 kcal/mol. Since $f(\varepsilon)$ is the only solvent variable a continuum model contains, the correlations requested are in fact one correlation, and only their slopes inform. Those slopes are chemically sensible: the solvent swings the reaction free energy by 129 kcal/mol, turning it exergonic above $\varepsilon \approx 8$, and displaces the TS earlier, leaving it *less* polarised in water than in vacuum. The model reproduces the direction and order of magnitude of the experimental effect (9.5 against 5.6 kcal/mol) but overestimates it, as expected from a method without electron correlation and a continuum without hydrogen bonds.

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